

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

November 2013

CL312 Environmental Engineering-I

Date: 25.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

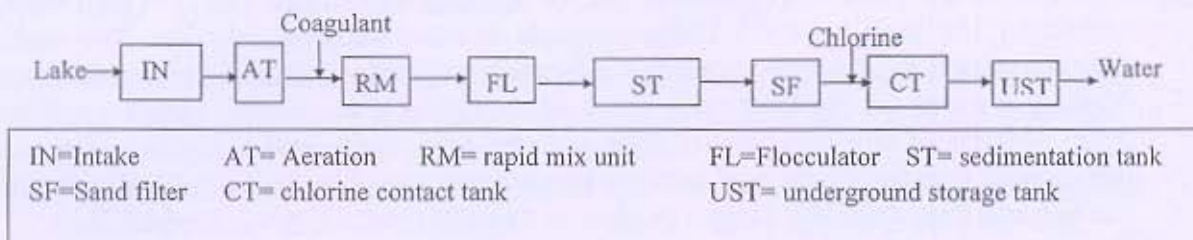
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Students can use self-made note of one A4 size page written on both sides.

SECTION - I

Q-1 The figure given below shows scheme of a water treatment plant. Answer following questions w.r.t. to this figure. (each answer=1 mark) [07]



1. The type of settling in sedimentation tank is _____. Select right choice and state why you selected it. (A) Type I (B) Type II (C) Type III (D) Type IV
2. Enlist the units which function based on physical forces (or enlist physical unit processes).
3. Enlist the units which work based on chemical forces (or enlist chemical unit processes).
4. Give at least two objectives (functions) of aeration.
5. Name the most widely used coagulant. Which parameter of water will change due to addition of coagulant?
6. If the sedimentation tank is of size 10 m (L) x 3 m (B) x 2.5 m (H) and receives a flow of 1200 m³/d, what is the surface overflow rate?
7. What will be the type of sand filter in this plant? RSF or SSF? Why?

Q-2 (a) A raw water sample has the following analysis: Na⁺ = 20 mg/L, Ca⁺² = 82 mg/L, Mg⁺² = 12 mg/L, Sr⁺² = 02 mg/L, Cl⁻ = 40 mg/L, SO₄⁻² = 72 mg/L, CO₃⁻² = 32 mg/L, HCO₃⁻ = 81 mg/L. Calculate hardness and total alkalinity in mg/L as CaCO₃. Also calculate carbonate hardness and non-carbonate hardness from above answers. Molecular weights (g/mole) are: Na=23, Ca=40, Mg=24, Sr=88, Cl=35.5, S=32, O=16, C=12, H=1. [07]

(b) Design a circular sedimentation tank to treat a water flow of 5 MLD. Assume suitable data. What size of particles of specific gravity 1.5 will be removed theoretically at 30 °C in this basin? [07]

OR

Q-2 (a) A rectangular sedimentation tank of size 25 m (L) x 6 m (B) x 3 m (H) is designed to treat a water flow of 9000 m³/d. A sieve analysis of suspended solids in water is: 60% solids are of size 0.05 mm, 25% solids are of size 0.03 mm, and 15% solids are of size 0.02 mm. If specific gravity of all solids is 2.65, determine % removal of suspended [07]

solids in this tank at 30 °C under ideal conditions.

- (b) For a flocculator unit treating 300 m³/h of water, determine volume of tank, power requirement, and size and number of paddles. Assume detention time = 20 min., average value of $G = 40 \text{ s}^{-1}$, paddle tip velocity = 0.6 m/s and relative velocity = 0.75 of paddle tip velocity, dynamic viscosity = $1.0087 \times 10^{-3} \text{ Pa.S}$, $CD = 1.8$. [07]

Q-3 (a) Explain briefly (any one): [02]

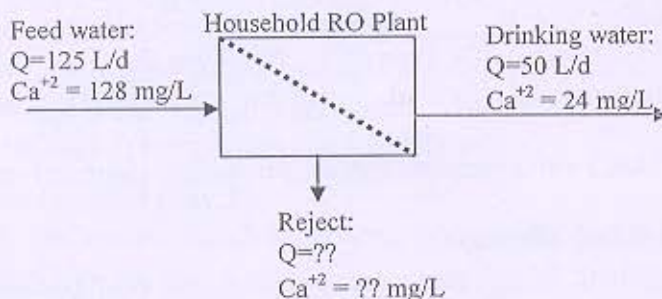
1. Schumtzdecky or dirty skin in SSF, 2. Optimum coagulant dose

(b) With the help of a neat diagram describe working of a rapid sand filter. [05]

(c) Design RSF for a town having population of 250000 with water demand of 180 LPCD. Assume rate of filtration as 5000 L/m²/h and L/B ratio of bed as 4/3. Also assume that 5% of filtered water is used for backwashing and each backwashing period is of 30 min. Find: (1) no. of filter beds, (2) quantity of washwater, (3) size and no. of washwater troughs if backwashing rate is 10L/m²/s. [07]

OR

Q-3 (a) A household RO plant daily produces 50L of drinking water from 125 L of feed water containing 128 mg/L of Ca^{+2} . Using principle of mass balance, calculate how much water is wasted as reject and what is Ca^{+2} concentration in it if Ca^{+2} concentration in drinking water is 24 mg/L. If there are 5 adults living in the house, each consuming equal volume of drinking water, what will be total Ca^{+2} intake of each. Doctors recommend that daily intake of calcium by an adult should be 800 mg. If all calcium has to come from drinking water only, do you think RO water is good for drinking? [04]



(b) A bed of uniform sand having particle diameter of 0.65 mm and specific gravity 2.65, porosity 0.42, and depth 65 cm is to be backwashed hydraulically. Calculate (a) backwash rate so that expansion of bed will be 50% of original depth, (b) headloss at this backwash rate. [05]

(c) Following chemical reaction takes place when ferric chloride is added as a coagulant in water. Calculate how much alkalinity as CaCO_3 will be consumed if coagulant dose is 20 mg/L? $\text{Fe}=56 \text{ g/mole}$. MW of other elements are given elsewhere in this paper. [05]

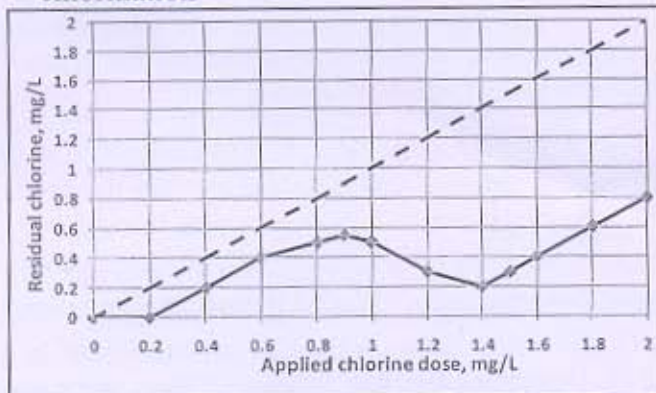
$$\text{FeCl}_3 + 3\text{HCO}_3^- \rightleftharpoons \text{Fe}(\text{OH})_3 \downarrow + 3\text{CO}_2 + 3\text{Cl}^-$$

SECTION – II

Q-4 Using the figure given below answer following questions.(each answer=1 mark) [07]

- What is this curve known as?
- What does the horizontal line between 0 mg/L and 0.2 mg/L on x axis indicate?
- At what applied chlorine dose the generation of free residual chlorine begins?
- What is the type of chlorine residues formed between applied chlorine dose of 0.2 mg/L and 0.9 mg/L?
- The chlorine demand at an applied chlorine dose of 1.6 mg/l is about

- (A) 1.6 mg/L (B) 0.4 mg/L (C) 1.2 mg/L (D) 2.2 mg/L
6. The applied chlorine dose needed to have a free available chlorine of 0.6 mg/L is
 (A) 0.6 mg/L (B) 1.8 mg/L (C) 0.9 mg/L (D) 0.4 mg/L
7. In a village, groundwater is chlorinated and then supplied to residents without any further treatment. This type of chlorination is called:
 (A) Super chlorination (B) Plain Chlorination (C) Prechlorination (D) Breakpoint chlorination



- Q-5 (a) [10] A town having population 50000 in 2013 decides to build a new intake in a nearby river to meet future demand of year 2035. As a Civil Engineer, you are asked to design an intake conduit (gravity main) using following data:
1. The population record of the town shows that population is increasing geometrically at 1.5% per year and will continue to increase for next 30 years at the same rate.
 2. Total water supplied to residents and commercial units in year 2013 was $3.5 \times 10^6 \text{ m}^3$. (hint: use this data to calculate LPCD).
 3. Water supplied to commercial units in 2013 was $3000 \text{ m}^3/\text{d}$. It is expected that this demand will be double by 2035.
 4. Assume that 5% of treated water is used in backwashing of filters.
- Calculate:
1. Total quantity of water the pipe has to carry (m^3/d).
 2. Diameter of an RCC pipe assuming velocity of flow = 1 m/s .
 3. Head loss using Hazen William's formula; value of $C=130$. Distance between intake and water treatment plant is 600 m.
- (b) [04] Some of the characteristics of a town which affect the per capita water demand are given below.

Parameter	Characteristics	Parameter	Characteristics
Size	Small	Quality of water	good
Community	Industrial	System of supply	Intermittent
Climate conditions	Hot	Billing of water use	By water meters
Standard of living	High	Water rates	High

From above table find out

- (1) Characteristics that tend to **increase** the water consumption & reason behind it.
- (2) Characteristics that tend to **decrease** the water consumption & reason behind it.

OR

- Q-5 (a) [09] At a town of present population of 50,000 with 160 lpcd water demand, water supply is provided through an over head tank. Water is pumped into this tank at a uniform rate continuously for 24 hrs of day. Length of supply main is 500m. The demand pattern is as shown in table.

Determine: (1) Daily demand in MLD, (2) Required capacity of overhead tank by analytical or graphical method, (3) Diameter of supply mains. Assume $f = 0.01$ & velocity through mains as 2 m/s.

Hours	Demand in % of 24 hour demand	Hours	Demand in % of 24 hour demand
0-8	5	16-20	25
8-10	40	20-24	10
10-16	20		

- (b) Water is conveyed from the source to a part of large Metropolitan city having flat ground profile & underground water source. Suggest and describe a water distribution system along with its type of layout for effective distribution of water throughout the city with minimum water losses. [05]

- Q-6 (a) Water is to be supplied to a town of 1.5 lakh population at the rate of 140 LPCD from a river, 1 km away. The difference in the elevation between the lowest water level in the intake and the service reservoir is 20 m. Determine the size of main. If the pumps are to be operated for total of 8h/day and the efficiency of pump is 80%, determine hp of pump required. Assume friction factor as 0.03 and velocity of flow through main = 1.5 m/s. Assume maximum demand = 1.8 times average demand of water. [07]

- (b) Draw a neat sketch of a house service connection and discuss functions of each component. [05]

- (c) Draw a typical house drainage plan of a residential building [02]

OR

- Q-6 (a) Why it is necessary to provide ventilation of the house drainage system? Draw a sketch to show the typical ventilation arrangements including inlet, outlet etc. to a house drainage system & explain its working. [04]

- (b) Do as asked (any 5): [10]

1. The correct match of Column-I (parameter) with Column-II (titrant used in the analysis) is:

Column I	Column II
P. Residual chlorine	1. 0.02 N Sulfuric acid
Q. Chloride	2. 0.01 N Sodium thiosulfate
R. Hardness	3. 0.0141 N Silver nitrate
S. Acidity	4. 0.01 M EDTA
	5. 0.02 N Sodium hydroxide

- (A) P-1, Q-2, R-4, S-1 (B) P-2, Q-3, R-4, S-1
(C) P-3, Q-2, R-4, S-5 (D) P-2, Q-3, R-4, S-5
2. Differentiate between True color and Apparent Color. (2-3 differences only)
3. Alkalinity in water is due to _____, _____, and _____. Which of these three predominates in natural water?
4. What is pseudo hardness? Answer in 2-3 sentences only.
5. Differentiate between combined and free residual chlorine. (2-3 differences only)
6. Why acidity, alkalinity, and hardness are expressed as CaCO_3 ? (2-3 sentences only)
